

AGENEWTECH

M3281V

Smart Module Specification_V2.2



Our aim is to provide customers with timely and comprehensive service. For any assistance, please contact our company headquarters:

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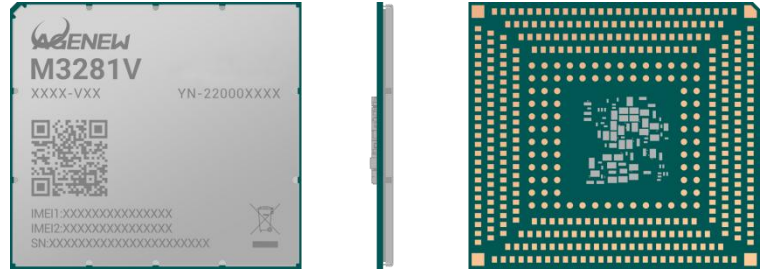
About the Document

History

Version	Date	Author	Description
-	2025-03-25	Daniel Wu	Initial
1.0	2025-03-27	Daniel Wu	Controlled
2.0	2025-12-18	Daniel Wu	Disable the WIFI and Bluetooth functions
2.1	2025-12-22	Daniel Wu	Update the memory capacity
2.2	2026-02-09	Daniel Wu	Update the company name

M3281V

Smart Computing Power Module



M3281V is a smart computing module based on MediaTek's Octa-core MT8893 4nm processor (1 × ARM Cortex-X4@3.25GHz + 3×ARM Cortex-X4@2.85GHz + 4×ARM Cortex-A720@2.0GHz) launched by AGENEW, with inbuilt ARM Immortalis-G720 GPU and 7th generation APU 790, equipped with Android 15 operating system. Its powerful performance, neon engine processing, 2D/3D graphics acceleration capabilities, fully meet the needs of customers for high-speed and high-computing-power functions in industrial and consumer applications.

M3281V has audio and video input/output interfaces and rich GPIO interfaces, including LCM display modules, touchscreens, cameras, microphones, speakers, UART, USB, I2C, SPI, PCIe, etc. It supports various audio and video codecs. The main screen supports up to WQHD (1600 × 3680)@180fps or 4K (2160 × 3840)@144fps. AI computing power reaches 48 TOPS.

M3281V is a SMT module, with a total of 544 pins, all of which are LGA pins, with a size of 42mm×46mm×2.6mm, which can be embedded in various M2M product applications through the pad, making it widely used in video conferencing, live streaming, gaming, edge computing, robotics, drones, augmented reality/virtual reality (AR/VR), and other terminal products and industries.



Main Advantages

- ✓ Built-in AI unit, support 48T computing power
- ✓ Support Android 15
- ✓ Support multiple audio and video codecs
- ✓ Main screen supports up to WQHD (1600×3680) @180fps or 4K(2160×3840)@144fps
- ✓ Support DP1.4
- ✓ Support 8K@30fps video recording and playback
- ✓ Abundant audio/video input and output interfaces and GPIO interfaces



1×ARM Cortex-X4@3.25GHz +
3×ARM Cortex-X4@2.85GHz +
4×ARM Cortex-A720@2.0GHz
64-bit Processor (Octa-core)



Support Android 15



48 TOPS



Powerful Multimedia

AI	M3281V
Region/Operator	China
Operating System	Android 15
Memory	128GB UFS + 16GB LPDDR5X
Module Dimension (mm)	42 × 46 × 2.6
Packaging	LGA
Temperature Range	
Operation Temperature	Operating temperature range: -20 °C ~ + 80 °C Storage temperature range: -40 °C ~ + 90 °C
Certification	
Compulsory Certification	Undetermined
Other Certifications	Undetermined
Features	
AI	48 TOPS
Video Encoding	8K@30fps
Video Decoding	8K@60fps
USB Firmware Update	•
OTA Firmware Update	•
Electrical Specification	
Operating Voltage	3.4~4.5 V, typical 3.8 V
Consumption (Typical)	200μA @ power off, 5.5mA @ sleep mode, typical
Interface	
LCD	2 × 4-lane MIPI DSI, max support WQHD (1600×3680)@180fps or 4K (2160×3840)@144fps
Camera	6 × 4-lane MIPI CSI,Support Single camera up to 320MP in non-zero mode or 108MP@30fps Support triple cameras (36MP + 36MP + 36MP)@30fps
TP	Capacitive touch screen, I2C control
Audio	Lineout Interface (External audio amplifier required) × 1, headphone × 1, simulated MIC ×6, digital MIC ×6
USB 2.0	USB2.0 × 2, USB 3.1 Gen2 × 1
UART	× 2
SD Card	× 1, support SD3.0, support 4-bit SDIO
ADC	× 6, universal ADC interface
I2C	× 17, for TP, Cameras, Sensors, etc.
SPI	× 4
PWM	× 1
Motor Driven	× 1
I2S	× 2
PWRKEY	Internally pull - up to VBAT
GPIO	× 20